

Neuroworks System Architecture

SYSTEM OVERVIEW

Neuroworks is an AI-powered agentic workforce platform that coordinates autonomous agents, shared tools, and governed workflows to complete complex work across the product.

Core Capabilities

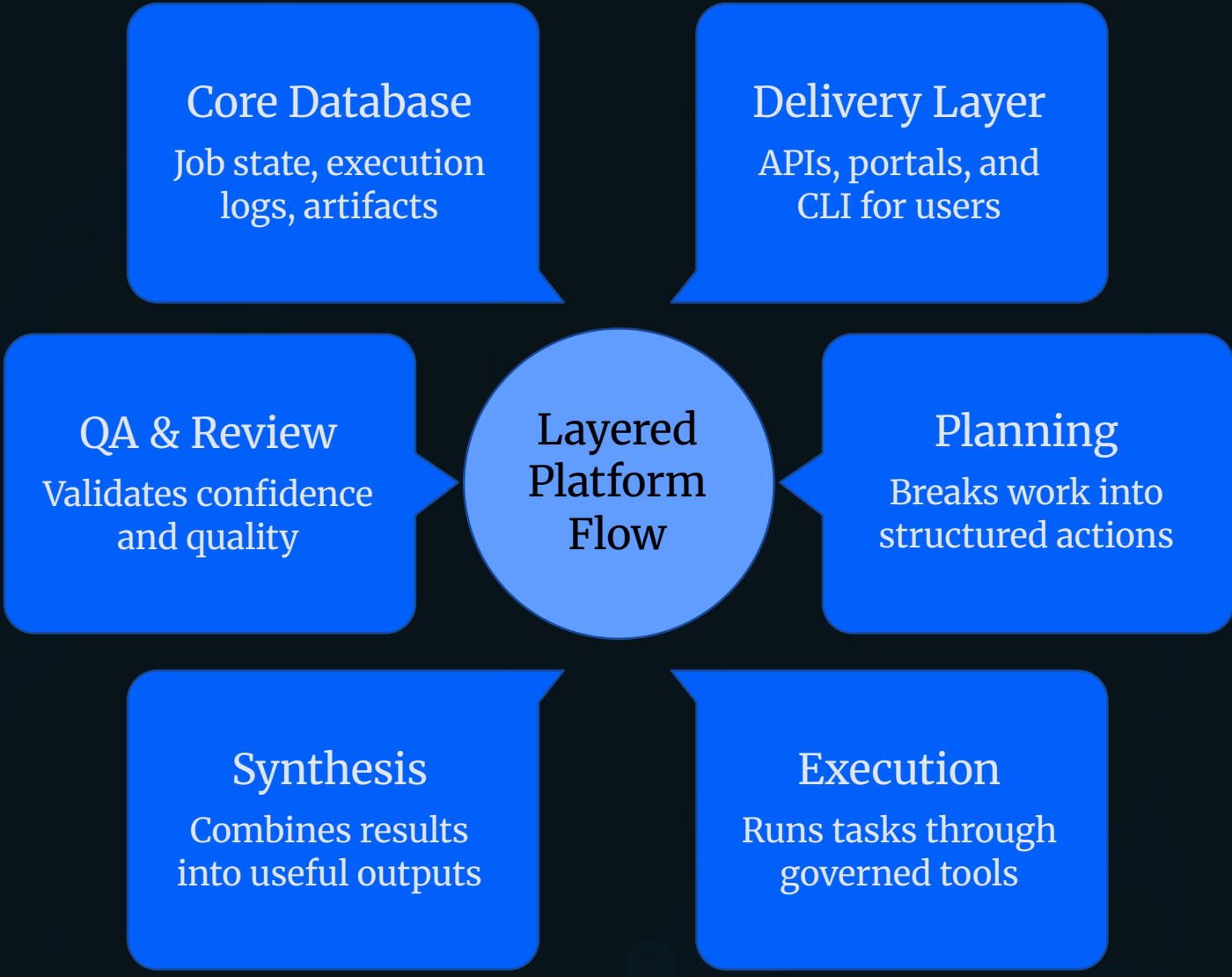
- Autonomous task execution across multi-step workflows
- Multi-agent orchestration with routing, handoff, and escalation
- Real-time monitoring of jobs, outputs, and system health
- Governed access to approved primitives and external capabilities
- Persistent execution tracking with logs, artifacts, and state history

Key Features

- Web UI for operators, reviewers, and admins
- API-first architecture for integrations and automation
- CLI-driven execution via Hermes for agentic control flows
- Structured outputs for reports, templates, and progress views
- Secure authentication, validation, and rate-limited access

HIGH-LEVEL ARCHITECTURE

The platform is organized as a layered pipeline that moves from user-facing delivery surfaces into orchestration, execution, and durable system storage.



Delivery surfaces initiate work, processing layers coordinate agent behavior, and the core database preserves state, traces, and generated artifacts for reliable operations and auditing.

Frontend Layer

React + Vite powers the operator experience and serves as the primary interface for managing work, reviewing progress, and inspecting results. During local development, `/api/*` requests are forwarded through the dev proxy to preserve same-origin behavior and simplify integration testing.

The UI supports task monitoring, report review, workflow inspection, and system configuration. Shared client modules organize resource access, handle token-based authentication through the Authorization header, and provide consistent request patterns across product areas.

Operators can interact with workforce tasks, review execution status, and change integrations without leaving the interface. The frontend is designed to keep the experience cohesive across user, admin, and reporting workflows.

Express Backend

The **openClaw Server** is the application backend that exposes the API surface for auth, chat, templates, executor jobs, connectors, payments, and governance. It acts as the central coordination point for routing requests into the platform’s operational domains.

Route groups are organized around product capabilities, while middleware enforces origin protection, schema validation, rate limiting, and structured error handling. This keeps the API observable, secure, and predictable under load.

The backend also supports the broader agent harness used by OpenClaw, providing the execution and control plane for the main pipeline.

Dual Execution Engines

Neuroworks uses two execution paths that work together: the main pipeline for standard processing, and Hermes for CLI-driven agent control and fallback handling when confidence drops or QA fails.



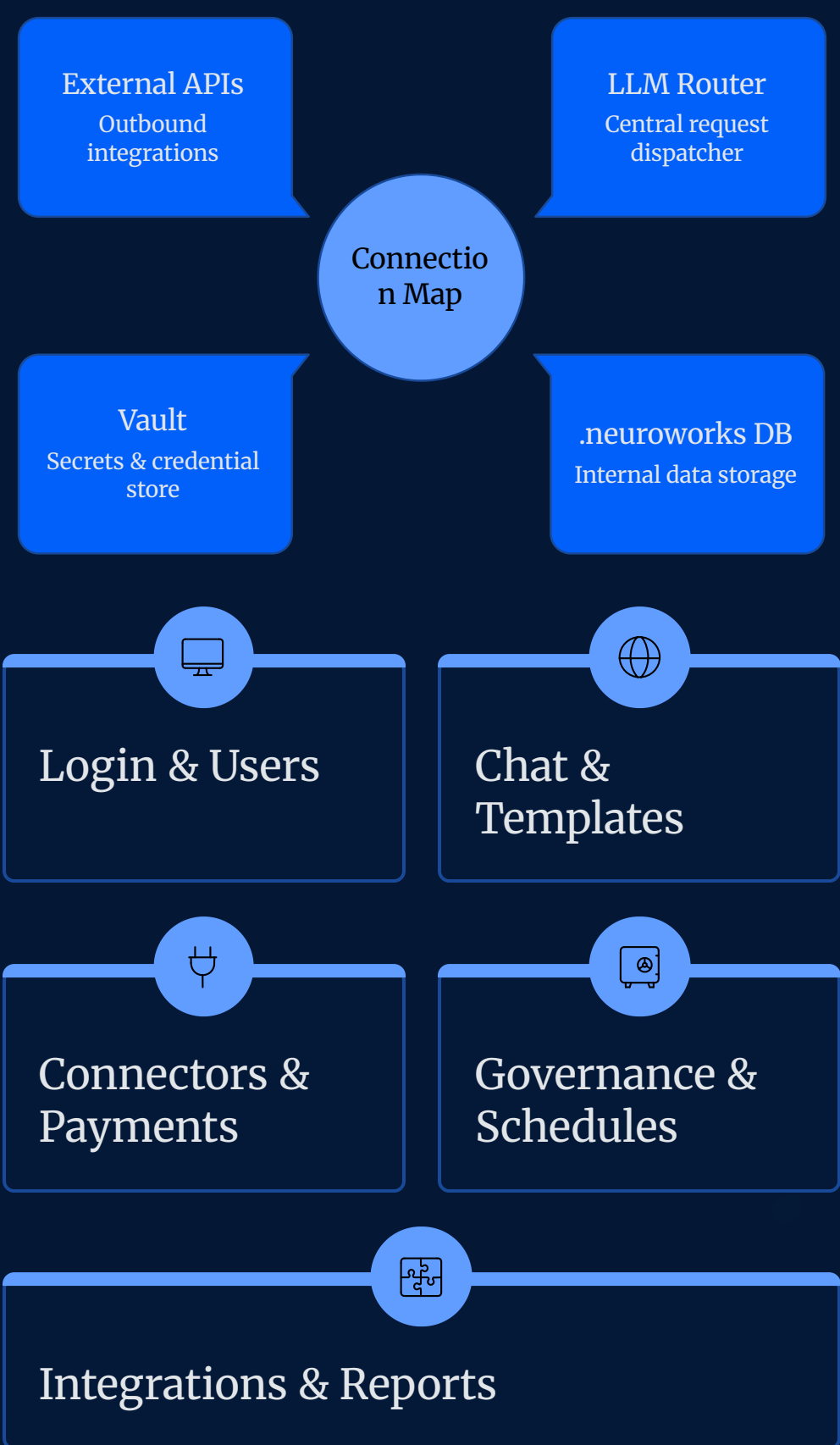
OpenClaw runs the main pipeline, while Hermes serves as a CLI agent that can take over, retry, or hand off execution when confidence falls below acceptable thresholds.

- Hermes calls approved capabilities through `/api/primitives/call` using allowlist-gated dispatch.

Primitives / Capabilities Layer



External Integrations & Storage



KEY OPERATIONAL FLOWS

Work moves through Neuroworks as a controlled sequence of intake, orchestration, execution, validation, and persistence. The platform keeps state synchronized so operators can inspect progress while agents continue processing in the background.

01

1. Intake

Requests arrive through the Web UI, APIs, or CLI and are authenticated before routing begins.

03

3. Execute & Synthesize

Agents act on tasks, gather results, and consolidate outputs into reports, actions, or follow-up steps.

02

2. Plan & Route

The system decomposes work into tasks, selects the right execution path, and assigns approved capabilities.

04

4. QA & Persist

Results are validated, stored as logs and artifacts, and fed back into system state for monitoring and review.

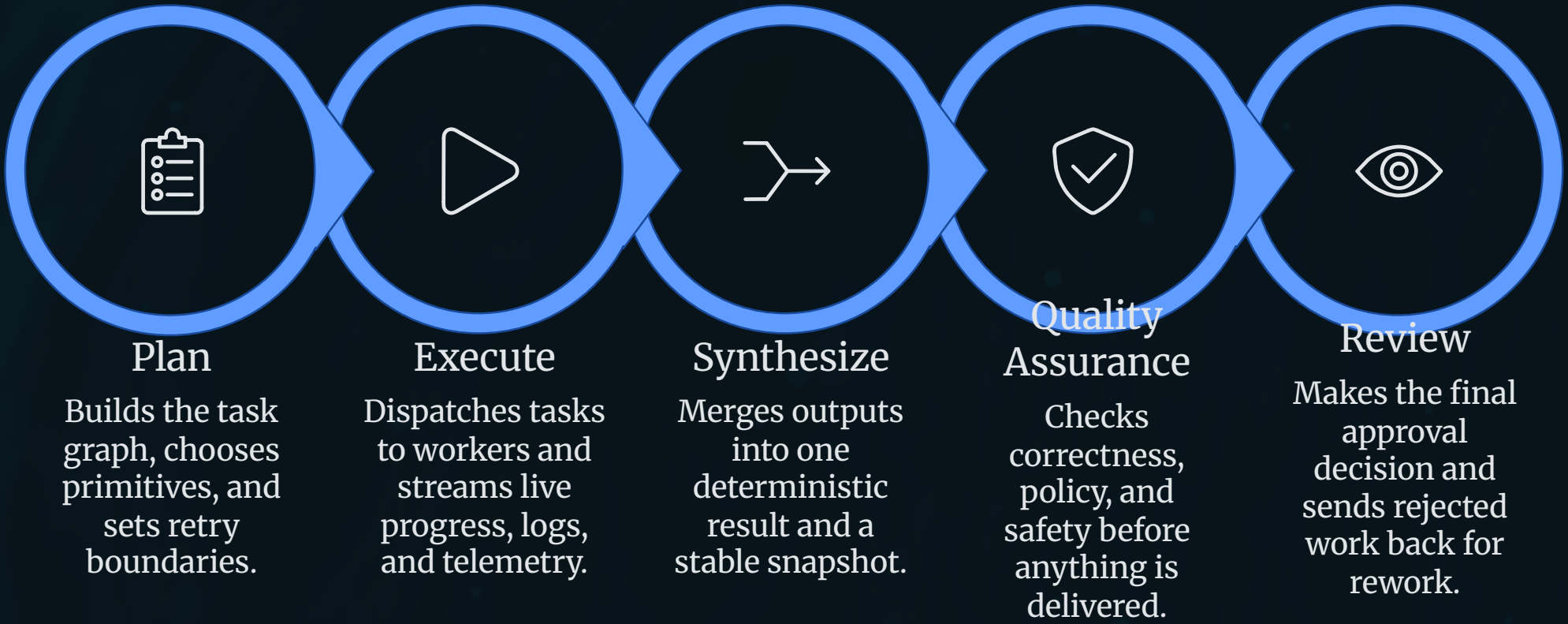
Neuroworks Pipeline: Execution Flow

SYSTEM OVERVIEW

Neuroworks runs as a closed-loop execution system: work moves from planning to execution, synthesis, QA, and review before delivery. State is preserved at every step so the pipeline can retry, recover, or hand off to Hermes Agent without losing progress.

PIPELINE STAGES

Each stage acts as a checkpoint in the loop, keeping work controlled, observable, and recoverable.



The loop stays resilient by checkpointing progress and allowing safe recovery when confidence drops or validation fails.

RETRY, RECOVERY & HANDOFF



Failures can retry locally, roll back to the last stable checkpoint, or hand off to Hermes Agent for controlled fallback execution.

MAIN PIPELINE VS HERMES FALLBACK



The main pipeline handles normal execution, while Hermes provides the fallback path when recovery or alternate handling is needed.

Agentic Agents: Autonomous Goal Pursuit

Agentic agents are advanced AI systems that operate autonomously. They perceive their environment, set goals, formulate plans, and execute actions with minimal human intervention, continuously learning and adapting based on feedback.

Perceive

Gather information and understand the current state of the environment.

Reflect

Evaluate outcomes, learn from experiences, and refine future plans and goals.



Plan

Formulate strategies and action sequences to achieve defined objectives.

Act

Execute the planned actions, interacting directly with the environment.

This iterative loop enables agents to handle complex tasks and navigate dynamic environments, improving their performance over time. This makes them ideal for roles requiring continuous adaptation and decision-making.